MACROECONOMIC DERIVATIVES: MORE VIABLE THAN FIRST THOUGHT!

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I. INTRODUCTION

Recently, Marshall, Bansal, Herbst and Tucker (1992) proposed a class of derivative products which they called macroeconomic swaps (macro swaps), and a related class of options which they called macroeconomic options (macro options).[1] These swaps and options have the potential to serve as the first effective tools for hedging business cycle risk.[2] Business cycle risk, also known as macroeconomic risk, refers to variations in a firm's performance measures that are associated with variations in the general level of economic activity.

The macro swap suggested by Marshall et al. is similar to other swaps in that one counterparty will pay a fixed price (or rate) on a quantity of notional principal in return for the second counterparty's paying a floating price (or rate) on the same quantity of notional principal.[3] What is different in the macro swap structure is that the floating leg is pegged to a macroeconomic index (or other measure) that varies directly with the general level of economic activity.

Bansal, Marshall, and Yuyuenyongwatana (1994) published findings from the first study on the viability of the macroeconomic derivatives concept. They chose as their macroeconomic indices two measures of consumer confidence—the Conference Board's Index of Consumer Confidence and the University of Michigan's Index of Consumer Sentiment. While there was no reason to believe that an index of consumer confidence should be the optimal index to which to peg the floating rate of a macroeconomic derivative product, the authors chose those indices because they are not subject to revision.[4]

Bansal et al. looked at the largest firms in 10 industries and measured the potential effectiveness of a macroeconomic swap hedge by measuring the correlation of each firm's revenues with each of the two consumer confidence indices. The Conference

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Board's Index (CBI) proved to have the greater explanatory power in all cases. Additionally, the analysis was repeated with different quarterly lags—in recognition that different firms' revenues might lag an index by different lengths. The authors also recognized the possibility of spurious correlation due to autocorrelation in the series—choosing to adjust for this problem using the Yule-Walker technique.

After adjusting for autocorrelation, the ability of the CBI to explain revenue volatility dropped precipitously in most cases. Indeed, of the 10 firms studied, the highest adjusted R^2 was 0.5864. This was for AT&T. However, as pointed out by the authors, the break-up of AT&T during the period covered by the data made the results suspect. The next highest R^2 was for IBM, which came in at 0.4647. The lowest was for Aetna Life and Casualty, which came in at 0.0700. Importantly, as expected, different numbers of quarterly lag were optimal for different firms. For Aetna the optimal lag was one calendar quarter. For IBM it was two quarters. For AT&T it was three quarters.

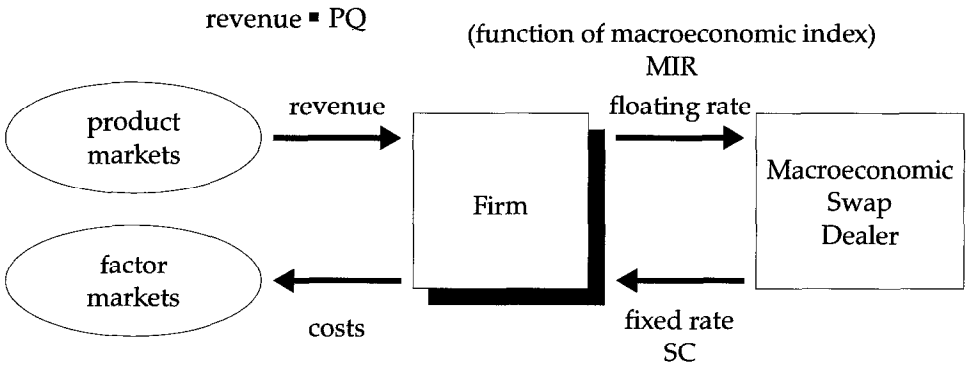
Macroeconomic derivatives have the potential to dwarf, in terms of transactional volumes of notionals, all other derivative products. Given the dimension of this potential, further study seemed warranted. This paper offers a second test of the macroeconomic derivatives (swaps) concept. It looks at the correlation of corporate revenues with gross domestic product (GDP). The viability of GDP as a macroeconomic index to which the floating leg of a swap might be pegged is demonstrated by the earlier finding that correlations are often maximized when revenues are lagged. *If all index revisions are complete before the swap's final settlement, then the revision issue is not the stumbling block it first appears to be.*

The same 10 industries, the same 10 firms, and the same revenue data used by Bansal et al. are employed in this paper in order that the results reported here might be directly compared to the earlier findings. It is shown that GDP is an excellent index to which to peg macroeconomic swaps. For several of the firms studied, the R^2s, even after adjusting for autocorrelation, exceeded 0.90. It is on this basis that we conclude that the potential for macroeconomic derivatives, as viable instruments for hedging business cycle risk, is actually considerably greater than suggested by earlier evidence.

The next section reviews the macro swap structure and some rudimentary hedging relationships that motivate the empirical analysis that follows. The third section explains the methodology utilized to assess the degree of explanatory power provided by GDP, the data employed, the lag structures incorporated, and the results obtained. The following section considers the use of regression betas to determine the size of the hedge that is appropriate for a given firm to manage its macroeconomic exposure. The fifth section offers a correction for spurious correlation induced by autocorrelation in the time series data.

II. THE MACROECONOMIC SWAP STRUCTURE AND HEDGE EFFECTIVENESS

As already noted, the "plain vanilla" macro swap structure is similar to any other basic swap structure in that there are two sides, called the fixed leg and the floating



cost = variable cost + fixed cost
= kQ + f

where P = price per unit
k = per unit variable cost
Q = quantity in units
F = fixed cost

floating rate: function of a macroeconomic variable that varies directly with the level of economic activity

(assumes that firm's sales are procyclical)

Figure 1. The Macroeconomic Swap Model

leg. The difference in these swaps is that the "rate" on the floating leg is pegged to a macroeconomic index or other measure that is known to vary directly with the general level of economic activity or, at least, the level of economic activity for the industry of which the firm is a part. Such swaps would be transacted, most likely, between a corporate (or government) end user and a macroeconomic swap dealer. This structure is illustrated in Figure 1.

Figure 1 depicts the firm's before-tax operating cash flow (OCF), equal to the difference between the operating revenue received from sales in its product markets and the operating cash expenses incurred in its factor markets. Variations in this operating cash flow, due to variations in revenue as a consequence of changing levels of economic activity, are the source of the risk that is to be hedged.[5] Assuming that the firm experiences procyclical OCF and that the macroeconomic variable to which the swap will be pegged varies directly with the firm's OCF, the swap would be structured to result in a net inflow to the firm if economic activity wanes and to result in a net outflow from the firm if economic activity heightens.

Hedge effectiveness, defined as the degree to which the variance of the firm's performance measure is reduced by the hedge, is routinely measured by the coefficient of determination (equivalent to the R^2 in a simple linear regression).[6] The risk remaining after the hedge has been placed is called basis risk. The relationship between basis risk and OCF variance is given by equation (1).

$$\textit{Basis risk} = (1 - R^2)VAR[OCF] \tag{1}$$

For simplicity of measurement, assume that there is a linear relationship between operating revenue and OCF.[7] This assumption allows the use of reported revenues, in lieu of OCF, in the analysis without any impact on the measure of hedge effectiveness. In other words, equation (2) is substituted for equation (1), and R^2 in equation (2) is the coefficient of determination between corporate sales revenue and the macroeconomic index to which the swap will be pegged.

$$Basis\ risk = (1 - R^2)VAR[REV] \tag{2}$$

It follows directly that the evaluation of alternative indices for pegging a macro swap should be evaluated on the basis of the R^2 obtained from either correlation or regression analyses.

III. CORPORATE REVENUES AND GROSS DOMESTIC PRODUCT

In this section, the impact of swings in GDP on corporate performance is examined. This is done by measuring the degree of correlation between GDP and corporate revenues. In order to assess the usefulness of this index as a predictor of fluctuations in corporate revenues, 10 major industries were selected for study. For each industry, the dominant firm, based on 1981 revenues, was employed in the study. (Holding companies that derived revenue from multiple sources were excluded from the study.) The industries and the dominant firms are provided in Table 1. In this section, no adjustment is made for autocorrelation in the data series. (This problem is addressed in Section V.)

A. Methodology

For each firm selected, quarterly revenue data covering the period January 1, 1981 through December 30, 1990 was obtained from Compustat. For each firm, the following linear regressions were performed:[8]

Table 1
Selected Industries and Firms

Industry	*Dominant Firm*	*TickerSymbol*
Pharmaceutical	Abbott Laboratories	ABT
Insurance	Aetna Life & Casualty Co.	AET
Banking	Chase Manhattan Bank	CMB
Financial Services	American Express	AXP
Telephone	AT&T	T
Aircraft	Boeing Company	BA
Health Care	Humana Inc.	HUM
Fast Food	McDonald Corp.	MCD
Computer	International Business Machines	IBM
Automobile	General Motors	GM

Table 2
Corporate Revenue vs. GDP Adjusted R^2s from the Regression

	$R_t = \alpha + \beta I_{t-s} + e_t$			
Firm	*S=0*	*S=1*	*S=2*	*S=3*
ABT	0.9497+	0.9517+	0.9502+	0.9489+
AET	0.5235^+	$0.500\ 6^+$	0.4838^+	0.4683^+
CMB	0.5729^+	0.5850^+	0.6065^+	0.6561^+
AXP	0.9509^+	0.9502^+	0.9502^+	0.9497^+
T	0.4636^+	0.5213^+	0.5546^+	0.5551^+
BA	0.7811^+	0.7909^+	0.8077^+	0.8115^+
HUM	0.9528^+	0.9571^+	0.9587^+	0.9598^+
MCD	0.9600^+	0.9590^+	0.9575^+	0.9566^+
IBM	0.6409^+	0.6270^+	0.6054^+	0.5787^+
GM	0.8708^+	0.8543^+	0.8468^+	0.8325^+
Mean	0.7666	0.7697	0.7721	0.7717
N	40	39	38	37

Note: $^+$significant at 0.1 percent.

$$R_t = \alpha + \beta I_{t-s} + \varepsilon t$$

where R_t denotes corporate revenue for quarter t, I denotes the index value (i.e., GDP), and s denotes the number of quarters of lag.

Optimal lag structures can, theoretically, be ascertained by employing more sophisticated statistical techniques. Such techniques, described by Hsiao (1981), have long been used. Nevertheless, simple single-period lags of lengths 0, 1, 2, and 3 calendar quarters were employed in this study because swap dealers would likely be limited to simple lag structures and single-periods of defined length in most swaps that they devise, particularly the early ones. The adjusted R^2s from the regressions are reported in Table 2.

B. Results

Aggregate predictive power is maximized with a lag of two quarters. With this lag, variations in GDP "explain," on average, 77.21 percent of the variations in corporate revenue. Notice, however, that in the aggregate the results from 0, 1, and 3 quarters of lag are not much different at 0.7666, 0.7697, and 0.7717, respectively. Nevertheless, certain of the individual R^2 were significantly different with different lags. For example, AET's R^2 was maximized with no lag (0.5235), that is, contemporaneous regression. For BA the R^2 was maximized at three quarters of lag. In every case, the R^2s were significant at 0.1 percent (0.001).

In addition to the degree of explanatory power evidenced by the regression R^2s, the direction of correlation is also of interest. The signs on the regression betas indicate that the correlations were positive in nine of the 10 cases. This confirms the expectation that most firms exhibit procyclical behavior with respect to gross domes-

tic product. One firm, AT&T, exhibited negative correlation, suggesting a countercyclical pattern to its revenue stream. However, AT&T should be considered a special case. During the period spanned by the data, this firm's industry underwent a transformation from a monopolistic structure to an oligopolistic structure. It was also impacted significantly by the spin-offs of its regional operating units in 1985.

IV. MAGNITUDE OF BUSINESS CYCLE RISK

While the R^2s provide evidence of the potential effectiveness of a macro swap hedge, they do not provide any evidence of the extent, in terms of dollar magnitudes, of the firms' exposures to this risk. This information, however, can be gleaned from the regression betas. The regression betas and accompanying *t*-scores, in parentheses, are reported in Table 3.

The magnitude of a regression beta provides information necessary to compute the size of the hedge needed to manage business cycle risk with macro swaps. The hedger would first determine the relationship between sales revenue and OCF (or profit). Then, with knowledge of the relationship between the index (e.g., GDP)

Table 3
Corporate Revenue vs. GDP Coefficients from the fRegression

	$R_t = \alpha + \beta I_{t-s} + e_t$			
FIRM	*S=0*	*S=1*	*S=2*	*S=3*
ABT	0.3655$^+$	0.3701$^+$	0.3768$^+$	0 .3829$^+$
	(27.149)	(27.388)	(26.586)	(25.874)
AET	0.6376$^+$	0.6319$^+$	0.6349$^+$	0.6391$^+$
	(6.622)	(6.252)	(5.973)	(5.7 20)
CMB	0.4285$^+$	0.4469$^+$	0.4740$^+$	0.5147$^+$
	(7.302)	(7.386)	(7.617)	(8.347)
AXP	1.883 3$^+$	1.9066$^+$	1.9432$^+$	1.9807$^+$
	(27.511)	(26.931)	(26.589)	(26.096)
T	–1.3199$^+$	–1.4447$^+$	–1.5532$^+$	-1.6202
	(–5.891)	(–6.511)	(–6.861)	(–6.776)
BA	1.5458$^+$	1.5953$^+$	1.6707$^+$	1.7217$^+$
	(11.840)	(12.030)	(12.508)	(12.489)
HUM	0 .3288$^+$	0.3347$^+$	0.3421$^+$	0.3505$^+$
	(28.064)	(29.145)	(29.505)	(29.322)
MCD	0.4213$^+$	0.4254$^+$	0.4343$^+$	0.4446$^+$
	(30.612)	(29.820)	(28.901)	(28.198)
IBM	3.7868$^+$	3 .7687$^+$	3.7535$^+$	3.7064$^+$
	(8.403)	(8.054)	(7.601)	(7.102)
GM	6.7967$^+$	6.8216$^+$	6.9331$^+$	6.9224$^+$
	(16.746)	(14.96 1)	(14.338)	(13.416)
N	40	39	38	37

Notes. Revenues measured in millions of dollars.
$^+$significant at 0.1 percent level.
T-statistics in parentheses.

and sales revenue, the hedger would determine the impact of fluctuations in the lag-adjusted index on OCF (or profit). Once the impact on OCF (or profit) has been determined, the hedger structures a macro swap pegged to the index, with the appropriate number of lags and the appropriate notional principal.

Let's consider a simple application. Suppose that the R^2 between a firm's revenue and the macroeconomic variable is 0.9598, as it is with HUM using three quarters of lag. Next, suppose that the regression results suggest that each one point decline in GDP is associated with a $0.3505 million decline in revenues, again, as it is with HUM with three periods of lag. Finally, suppose that there is a linear relationship between revenue and OCF such that each one point decline in GDP translates to a decline in OCF of $0.1200 million. The swap would then have to be structured in such a way that the net cash flow to HUM increases (decreases) by $0.1200 million for each 1 point drop (rise) in GDP. This is merely a matter of properly specifying the reference rate, relative to its initial value, and the notional principal on the swap.

V. ADJUSTING FOR AUTOCORRELATION

The presence of autocorrelation and nonstationarity in the data series can introduce spurious correlation and bias the estimate of hedge effectiveness. For this reason, portions of the analysis were repeated with a correction for autocorrelation. Procedures to correct for these biases are discussed in Hendry (1986), Box and Jenkins (1976), Judge, Griffiths, Hill, and Lee (1985), Gallant and Goebel (1978), and Harvey (1981).

The Durbin-Watson statistic was used to test for autocorrelation in the index and the revenue series. Some autocorrelation was found, but not in all the series. This contrasts with the results of Bansal et al., where autocorrelation was found in every

Table 4
Corporate Revenue vs. GDP Adjusted R^2s from the Yule-Walker Estimates

	S=1		*S=2*		*S=3*	
Firm	*Raw*	*Y-W*	*Raw*	*Y-W*	*Raw*	*Y-W*
ABT	0.9517	0.9241	0.9502	0.9231	0.9489	0.9217
AET	0.5006	0.1797	0.4828	0.1723	0.4683	0.1679
CMB	0.5850	0.2234	0.6065	0.2539	0.6561	0.3094
AXP	0.9502	0.9062	0.9502	0.9115	0.9497	0.9123
T	0.5213	0.2873	0.5546	0.3272	0.5551	0.3294
BA	0.7 909	0.7243	0.8077	0.7576	0.8115	0.7748
HUM	0.9571	0.8728	0.9587	0.8766	0.9598	0.8727
MCD	0.9590	0.9273	0.9575	0.9283	0.9566	0.9281
IBM	0.6270	0.7991	0.6054	0.7868	0.5787	0.7678
GM	0.8543	0.8050	0.8468	0.7789	0.8325	0.7462

Notes: Raw values refer to the R^2s from regressions not corrected for autocorrelation. Y-W refers to those that were. The Yule-Walker (Y-W) technique does not provide levels of significance for the R^2s that it generates.

series. To adjust for the presence of autocorrelation, the two-step full transform method, which employs the Yule-Walker estimation[9], was used to minimize the problem. The Yule-Walker is the default autoregressive method of the Statistical Analysis System (SAS).

The regression R^2s from that analysis are reported in Table 4 for lags of 1, 2, and 3 quarters and contrasted with the raw adjusted R^2s from Table 2.

Notice that the R^2s are not meaningfully impacted, in most cases, by the adjustment for autocorrelation. This suggests that the benefits a macroeconomic hedge would offer are real, not illusory.

VI. SUMMARY AND CONCLUSIONS

This study has provided evidence on the effectiveness of macroeconomic swaps to hedge business cycle risk when the floating rate on the macro swaps is pegged to gross domestic product. The choice of GDP seems more appropriate than measures of consumer confidence that were employed in an earlier study. Nevertheless, it seems reasonable to conclude that different indices of macroeconomic activity might be appropriate for different firms.

The high correlations found for many of the firms studied, even after adjusting for autocorrelation, suggest that macroeconomic derivatives have considerable potential to offer a viable vehicle to hedge a firm against otherwise uncontrollable business cycle risk.

NOTES

1. Just as a macroeconomic swap is an extension of the basic swap model, a macroeconomic option is an extension of the basic option model, i.e., macro options are analogous to multi-period caps and floors except that the reference rate is a macroeconomic index rather than an interest rate index. Our focus in this paper is on the macroeconomic swap, but the arguments are easily extended to macroeconomic options. For a comprehensive discussion of multi-period options, see Marshall and Bansal (1992, 1994, Ch. 15). Macroeconomic derivatives are also discussed in Marshall and Kapner (1993, Ch. 9).
2. Hedging business cycle risk with macroeconomic derivatives was the focus of a session at the 1992 annual meeting of the International Association of Financial Engineers.
3. For a fuller discussion of swap structures and uses as well as a detailed review of the swaps literature, see Abken (1991), Marshall and Kapner (1993), Schwartz and Smith (1990).
4. Index revision could create a serious settlement problem for the counterparties in swap should the revision occur after the payment of the settlement sums.
5. Ideally, it is before-tax profit that we wish to hedge, not operating cash flow. However, as outside analysts, our available data is limited to those income

statement components reported by the corporations we studied. This data is often massaged during periods of poor profit performance by resort to various window dressing tactics including debt-equity swaps, debt defeasance, inventory revaluations, and extraordinary profits from the sale of assets. Thus, operating cash flow is a purer measure of reported performance. Purer still is sales revenue, as this is the most difficult corporate performance measure to distort. Clearly, however, corporate managers who have access to historic data stripped of these distortions, should focus their hedging efforts on before-tax profit.

6. The coefficient of determination has served as the standard measure of hedge effectiveness since it was first proposed by Johnson (1960). Other contributions to the hedging literature that have built on this relationship include the work of Stein (1961) and Ederington (1979). Many of the statistical anomalies that we note but do not address in this paper have also been considered in the context of hedging theory. Much of this literature is summarized in Herbst, Kare, and Marshall (1993).
7. As already noted, revenue has the advantage of being less easily manipulated than either profits or cash flows. Employing reported sales revenue, in lieu of profit or cash flow, is a much simpler approach and produces an identical result if we assume a linear relationship between revenue and operating cash flows. Of course, to the degree that this relationship is other than perfectly linear, some distortion will result. This problem, however, can be alleviated in actual applications of these instruments by focusing on profit—the true target of the hedge—rather than on revenue.
8. The regressions were performed on levels rather than on differences. Differences are, generally, the preferred variable—to correct for aberrations induced by changing levels over the period of study. This was not done because later in the paper, we report results incorporating the Yule-Walker technique which provides an alternative adjustment for this problem.
9. Harvey and McAvinchey (1978) suggest that the Yule-Walker two step transformation is just as efficient as the maximum-likelihood method but is computationally more efficient.

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